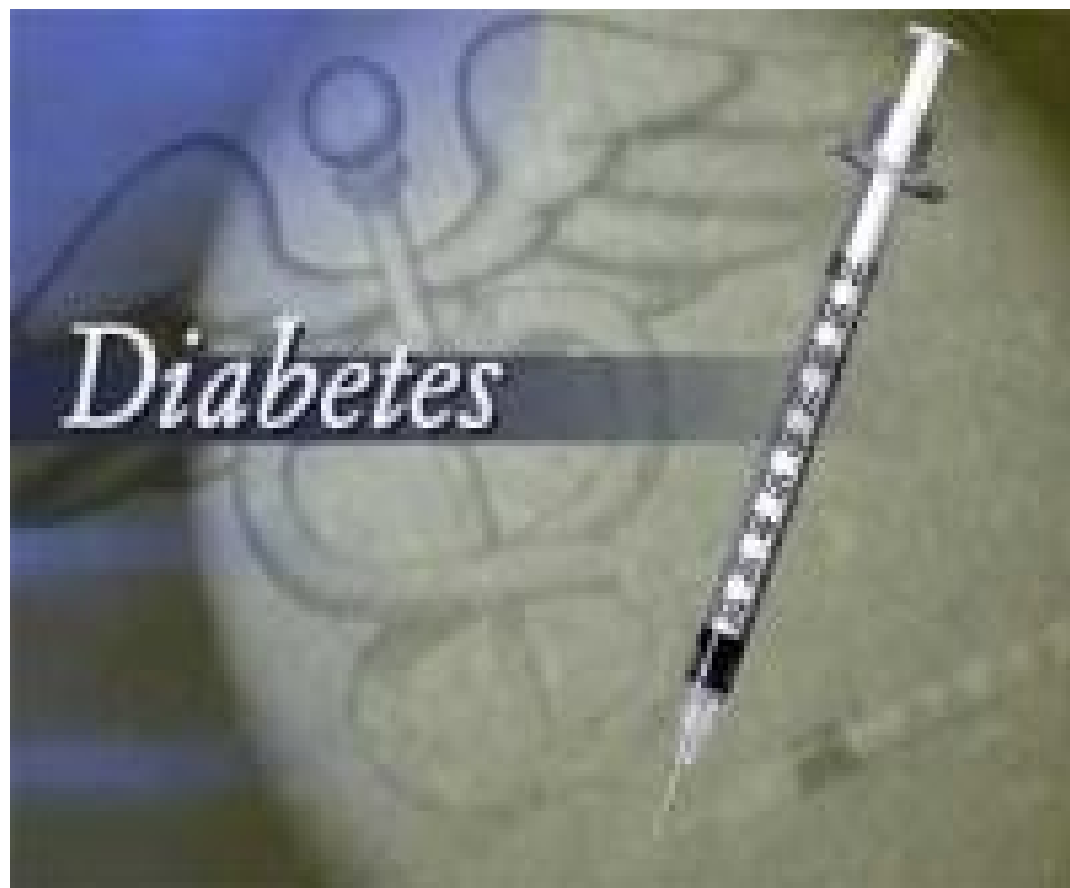




What is Diabetes Mellitus?

Normal Glucose Metabolism:

- When the amount of glucose in the blood increases, e.g., after a meal, it triggers the release of the **peptide hormone insulin** from the pancreas.
- Insulin **stimulates muscle and fat cells** to take-in glucose from the blood and stimulates the liver to metabolize glucose, causing the blood sugar level to decrease to normal levels.

Diabetes Mellitus -

- **Diabetes mellitus is a metabolic disorder characterized by abnormally high levels of sugar (glucose) in the blood.**
- **Condition of high levels of glucose in the blood is called Hyperglycemia.**
- **Hereditary and environmental factors are responsible.**

Causes of Hyperglycemia

- **Insulin** is a hormone secreted from the **Islet cells** of the **Pancreas**.
- **Insulin enables cells** to **absorb glucose** in order to turn it into energy, thereby keeping the normal blood glucose level in the blood.
- In diabetes, blood **sugar levels remain high**, may be because insulin
 - is **not being produced** at all
 - is **not made at sufficient levels**, or
 - is **not as effective** as it should be.

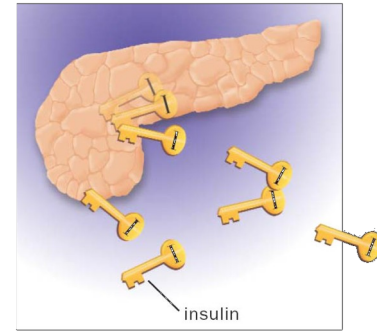
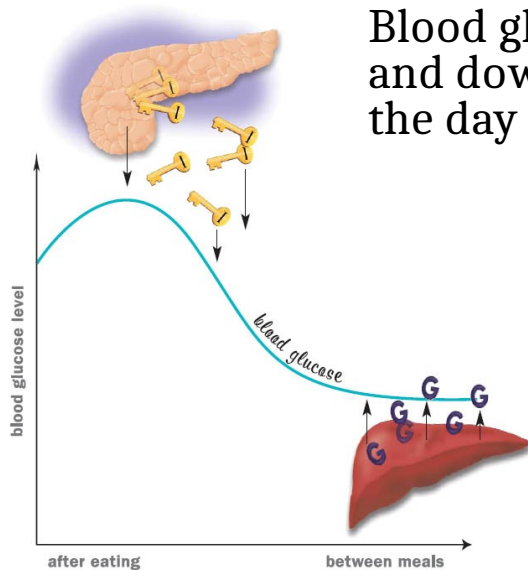
This causes glucose to accumulate in the blood, often leading to various complications.

When you eat, your body breaks food down into glucose. **Glucose** is a type of sugar that is your body's main source of energy.

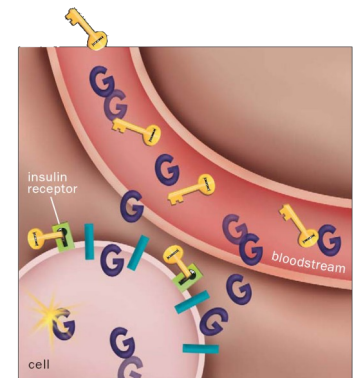
As blood glucose rises, the body sends a signal to the pancreas, which releases **insulin**.

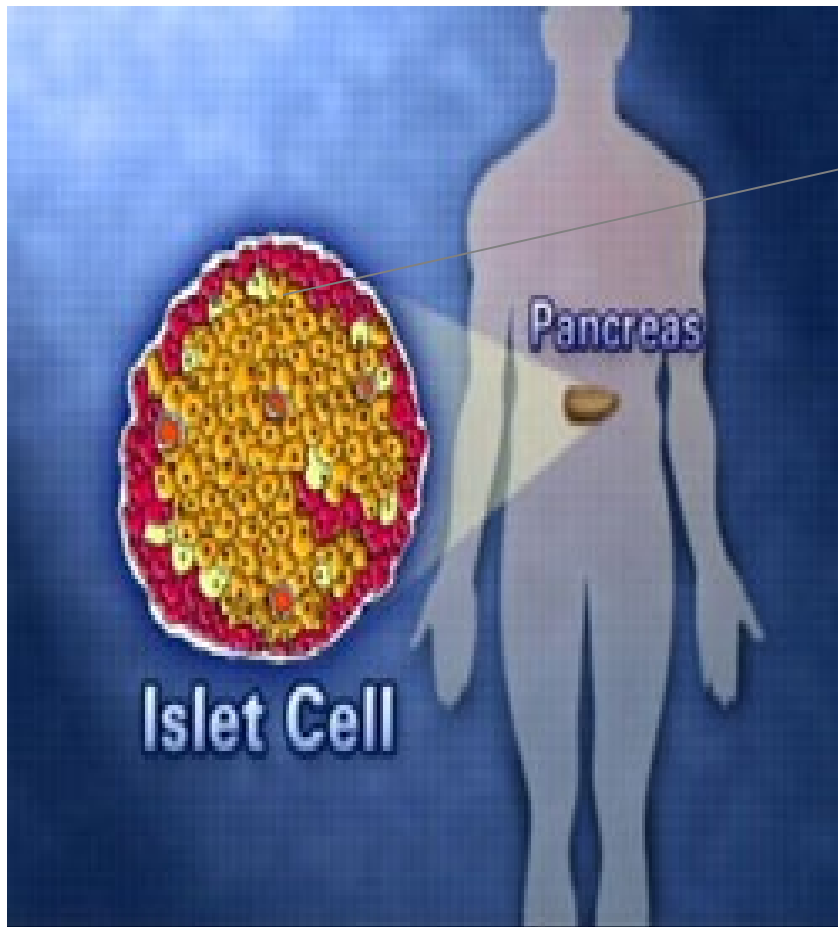
As your blood glucose **rises** (after a meal), the pancreas releases insulin.

Blood glucose goes up and down throughout the day

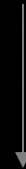


Acting as a key, insulin binds to a place on the cell membrane (an **insulin receptor**), unlocking the cell so glucose can pass into it. There, most of the glucose is used for energy right away.

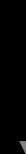




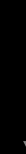
If insufficient/lack of insulin



Glucose can't enter the cell



Glucose remains in blood



Diabetes Mellitus/Hyperglycemia

Types of Diabetes

There are mainly three types of diabetes:

1. Type I or insulin dependent diabetes often called **juvenile onset** diabetes,
 2. Type II or non-insulin dependent diabetes often called **adult onset** diabetes &
 3. Gestational diabetes (GDM) – during pregnancy
- The most common forms of diabetes are type 1 diabetes (5%), which is an autoimmune disorder and type 2 diabetes (95%), which is associated with obesity.
 - Gestational diabetes is a form of diabetes that occurs during pregnancy.

Type 1 diabetes:

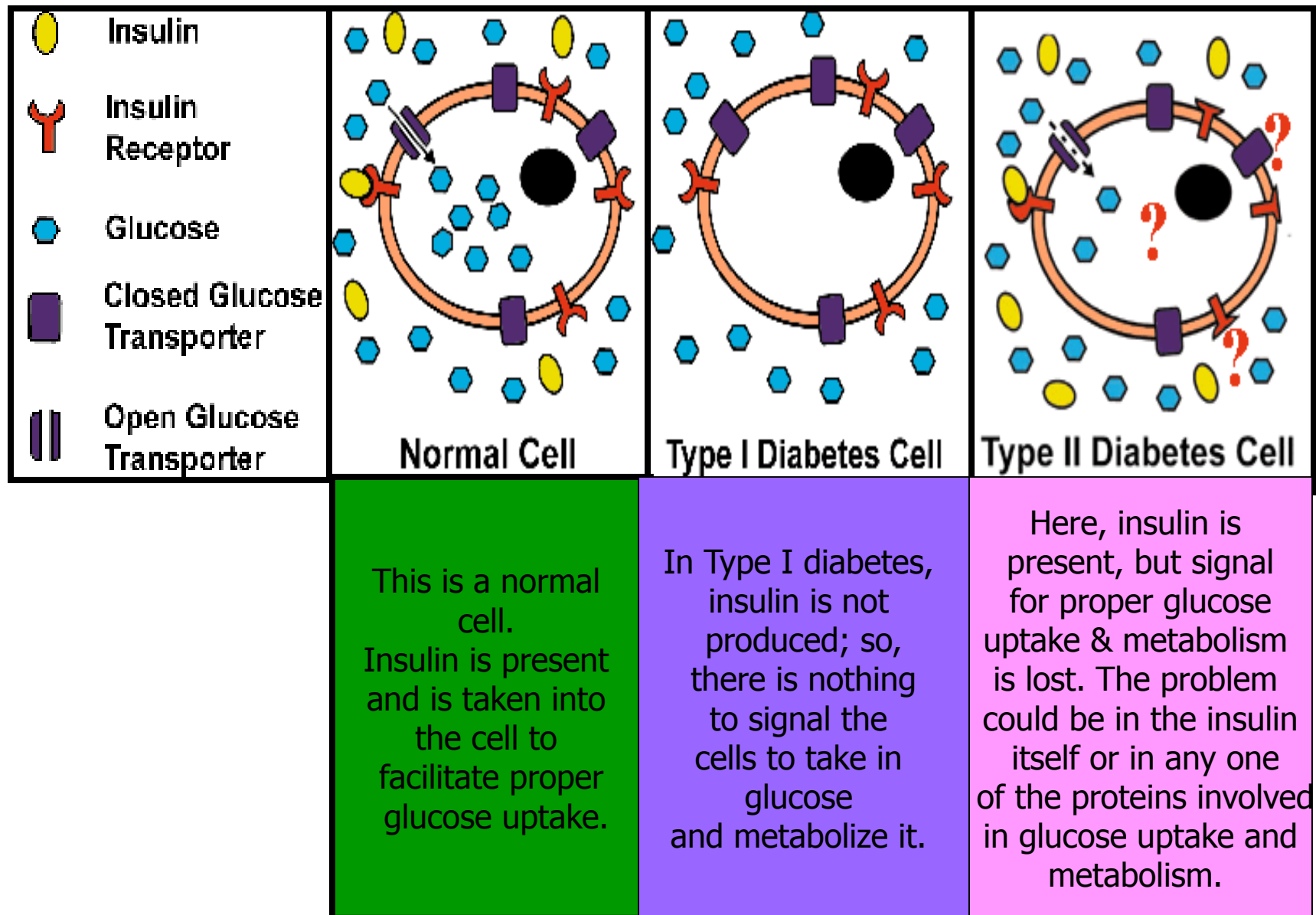
- Most commonly occurs in children
- Is a result of the body's immune system attacking and destroying the beta islet cells.
- The trigger for this autoimmune attack is not clear, but the result is the end of insulin production.

Type 2 diabetes:

- Type 2 diabetes commonly occurs in adults who are obese.
- There are many underlying factors that contribute to the high blood glucose levels in these individuals.
- An important factor is the body's resistance to insulin, essentially ignoring its insulin secretions.
- A second factor is the falling production of insulin by the beta cells of the pancreas.

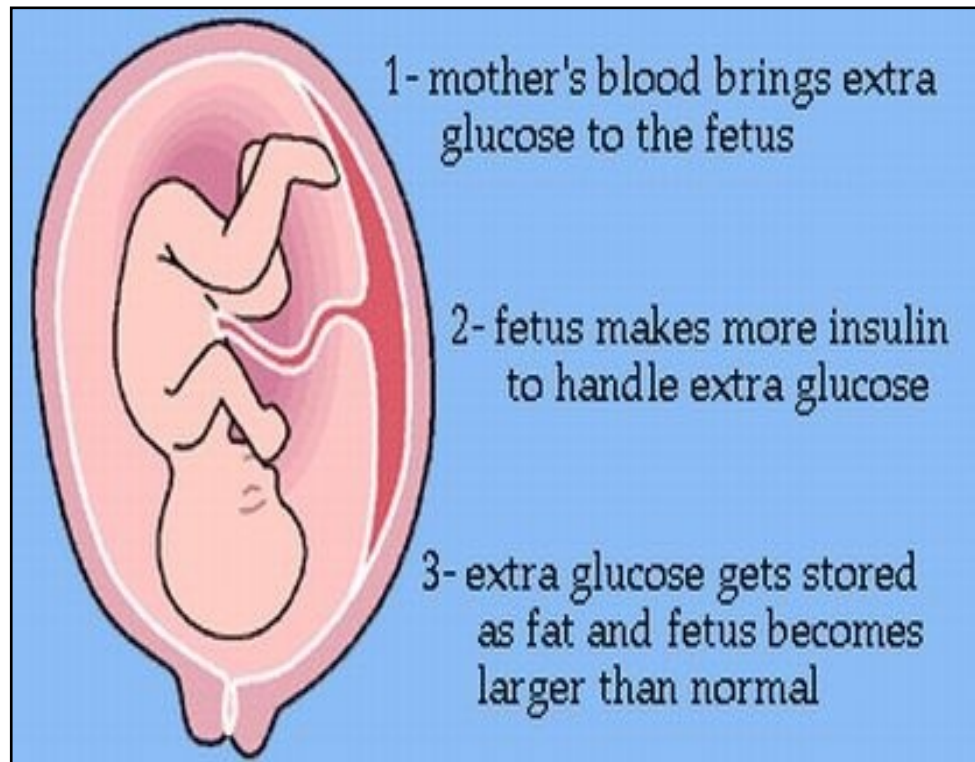
Therefore, an individual with type 2 diabetes may have a combination of deficient secretion and deficient action of insulin.

Type I & II Diabetes : Insulin Response



Gestational Diabetes- 3rd Type of Diabetes

Gestational diabetes is diabetes that happens for the first time when a woman is pregnant. Gestational diabetes goes away when any woman has her baby, but it increases her risk for having diabetes later.



Due to Gestational Diabetes, the baby has risks for-

- 1) **Macrosomia** - refers to a baby that is considerably larger than normal.
- 2) **Birth injury** - may occur due to the baby's large size and difficulty being born.
- 3) **Hypoglycemia** – refers to low blood sugar in the baby immediately after delivery.
- 4) **Respiratory distress**- difficulty in breathing.

Signs & Symptoms

HIGH BLOOD SUGAR *Hyperglycemia*

Signs and Symptoms:



Dry Mouth



Extreme Thirst



Frequent Urination



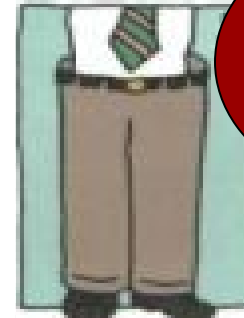
Always Tired



Frequent Bed wetting

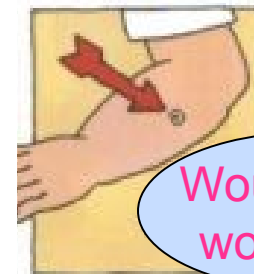


Stomach Pain



Sudden
Weight
Loss

Always
Hungry



Wounds that
won't heal

Diagnosis

Diabetes is diagnosed by measuring the blood glucose level during fasting (**Fasting plasma glucose test**) or an **oral glucose tolerance test (OGTT)**.

Three status associated with diabetes: **people who are not diabetic**; those with **Impaired Glucose Tolerance (IGT)**; and those with **diabetes**.

Category	Blood Glucose (mg/dL)	
	Fasting	After meal
Normal	<115	<140
Diabetic	>140	>200
Impaired Glucose Tolerance	<140	140-199

Good News for Type 1 Diabetes

Nephropathy

Neuropathy

Retinopathy

Keeping blood glucose in target range reduces:

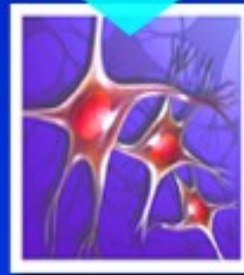
Kidney disease

**as
much as
56%**



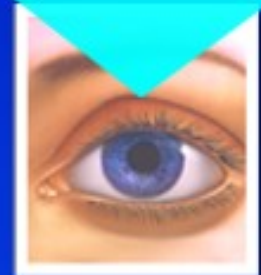
Nerve damage

**as
much as
60%**



Eye disease

**as
much as
76%**



Lipid profile

What is a lipid profile?

A **lipid profile or lipid panel** is a panel of blood tests used to find **abnormalities in blood lipid** (such as cholesterol and triglycerides) concentrations. The lipid profile is a group of tests that are often ordered together to determine risk of coronary heart disease. The tests that make up a lipid profile are tests that have been shown to be good indicators of whether someone is **likely to have a heart attack or stroke** caused by blockage of blood vessels (**hardening of the arteries**), can **identify certain genetic diseases**, certain forms of **pancreatitis**, and **other diseases**.

looking for information about the amounts of four types of fats in the blood.

What tests are included in a lipid profile?

The lipid profile includes:

Total cholesterol,

HDL-cholesterol (often called good cholesterol),

LDL-cholesterol (often called bad cholesterol),

VLDL-cholesterol (This type of cholesterol carries triglycerides)
and

Triglycerides (TG).

Sometimes the report will include additional calculated values such as the Cholesterol:HDL ratio or a risk score based on lipid profile results, age, sex, and other risk factors.

A lipid panel measures **five different types** of lipids from a blood sample, including:

- **Total cholesterol:** This is your overall cholesterol level — the combination of LDL-C, VLDL-C and HDL-C.
- **Low-density lipoprotein (LDL) cholesterol:** This is the type of cholesterol that's known as “bad cholesterol.” It can collect in your blood vessels and increase your risk of cardiovascular disease.
- **Very low-density lipoprotein (VLDL) cholesterol:** This is a type of cholesterol that's usually present in very low amounts when the blood sample is a fasting sample since it's mostly comes from food you've recently eaten. An increase in this type of cholesterol in a fasting sample may be a sign of abnormal lipid metabolism.
- **High-density lipoprotein (HDL) cholesterol:** This is the type of cholesterol that's known as “good cholesterol.” It helps decrease the build-up of LDL in your blood vessels.
- **Triglycerides:** This is a type of fat from the food we eat. Excess amounts of triglycerides in your blood are associated with cardiovascular disease and pancreatic inflammation.

Cholesterol helps the body form hormones, vitamin D and other important substances, but too much of it in the blood can clog and damage the blood vessels. Because it is a fat-like substance that doesn't mix with blood, cholesterol has to combine with proteins to form lipoproteins. Lipoproteins can travel in the blood to all the organs and tissues of the body.

Low-density lipoproteins (LDLs, or "bad" cholesterol) build up in the blood and increase your risk of heart disease.

High-density lipoproteins (HDLs, or "good" cholesterol) carry cholesterol to the liver, where it is removed from the body.

Triglycerides store energy for your body to use when it is needed. If there is too much, it can block blood vessels and cause other health problems such as abdominal pain and pancreatitis.

How is a lipid profile used?

The lipid profile is used to guide providers in deciding how a person at risk should be treated. The results of the lipid profile are considered along with other known risk factors of heart disease to develop a plan of treatment and follow-up.

It's important to know the ranges....

- **Total Cholesterol** (lower is better)

- Best = <200 mg/dL
- Borderline high = 200-239 mg/dL
- High = 240 mg/dL or higher

- **Triglycerides** (lower is better)

- Best = <150 mg/dL
- Borderline high = 150-199 mg/dL
- High = 200-499 mg/dL
- Very high = 500 mg/dl or higher

- **LDL Cholesterol** (lower is better)

- Best = <100 mg/dL
- Good = 100-129 mg/dL
- Borderline high = 130-159 mg/dL
- High = 160-189 mg/dL
- Very high = 190 mg/dL or higher

- **HDL Cholesterol** (higher is better)

- Low = <40 mg/dL
- Best = 60 mg/dL or higher

What is cholesterol ratio?

Cholesterol ratio (CR) is calculated by **dividing** your **total cholesterol** by the **amount of high-density lipoproteins (HDL)** in your blood. It is recommended that your cholesterol ratio stays **under 5**, with the **ideal value being 3.5**. Essentially, as your cholesterol ratio increases, so does the risk of you getting a heart disease.

$$\text{CR} = \text{TC} \div \text{HDL}$$

A desirable ratio is 5:1, and an optimal ratio is 3.5:1. The lower this number is, the healthier a person's cholesterol levels are.

Value	Rating
2.3	Good
2.9	Good
3.5	Good
4.0	Borderline high
4.7	Borderline high
5.3	High
5.9	Very High

Cholesterol within VLDL particles are reported to be contained with triglycerides in a ratio of [triglycerides]:[cholesterol] = 5 (for mg/dL unit) or 2.26 (for mmol/L)

$$\text{VLDL-C} = [\text{Triglycerides}] / 5$$

$$\text{LDL-C (mg/dL or mmol/L)} = [\text{Total cholesterol}] - [\text{HDL-C}] - [\text{Triglycerides}] / 5 \text{ or } 2.26$$